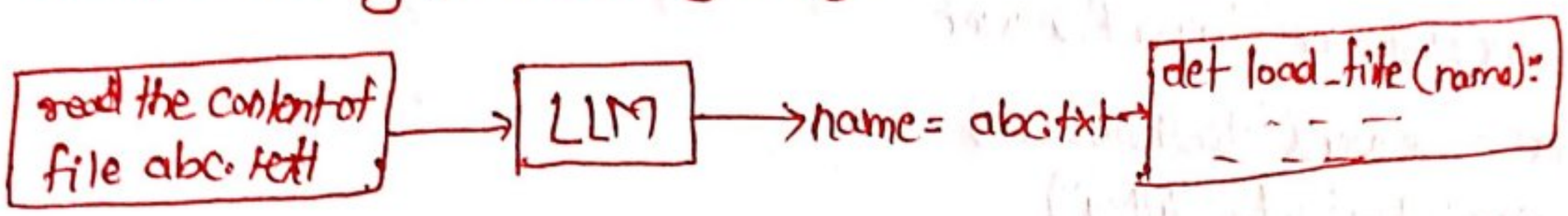
MCP * Model Context Protocol

What is Context

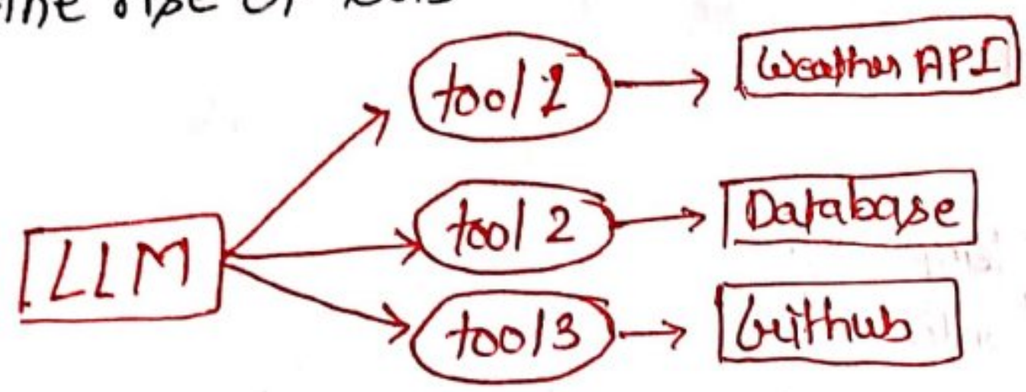
- Context is everything an AI can "see" when it generates a response.
- Context refers to the information (conversation, history) that the LLM used to generate a response.

function calling.

- Function calling is the way using which LLM can call ext functions.



The rise of tools



Implication of tools

- Salesforce integrations for sales teams
- Slack bots that could read msg history and channel context.
- Google Drive connectors for document access and collaboration.
- Database query tools that could analyze company data.
- Github integrations for code review and pull request management.
- Cursor - builds filesystem access and intelligent code search.
- Perplexity - add web browsing and real time information retrieval
- ChatGPT Plus - browsing, file uploads, code ~~edition~~ execute
- Claude - gained computer use capabilities.

Problem with Tools.

- Integration Problem

  no. of bots x no. of tools

- Maintenance problem
- Security fragmentation
- Cost & time wastage.

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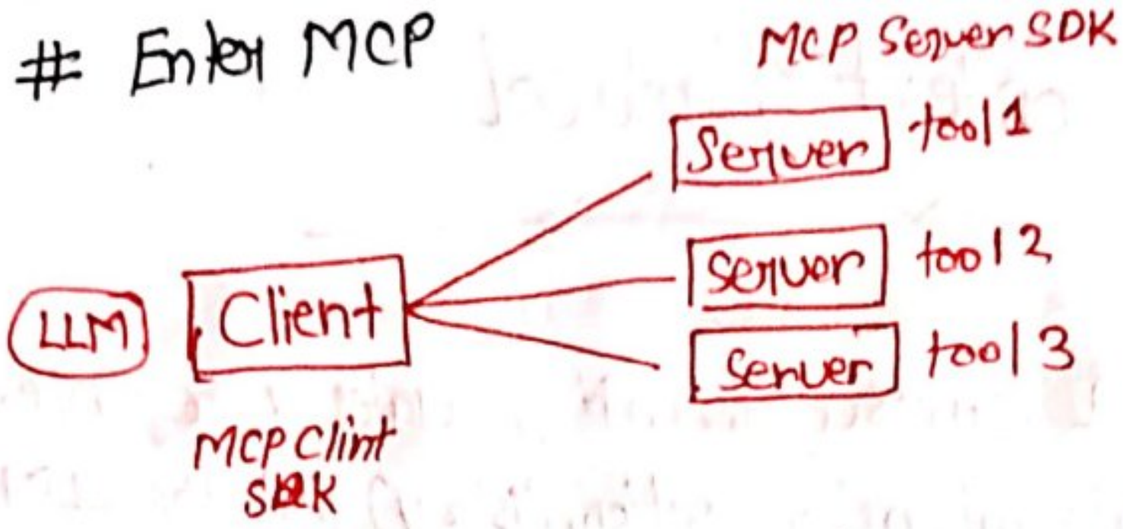
Enter MCP



MCP code for ~~weather~~ weather

- Client

result = callTool("get_weather", {"city": London"})

- Server

```
from mcp.server. import Server
Server = Server("WetherServer)
@ Server.tool("get_weather")
def get_weather(city: str):
    data = lookup_weather(city)
    return. {
        "temperature": data["temp_c"],
        "Condition": data["Condition"]
    }
Server.start()
```

Server does the Reavy lifting

- authentication with github
- API rate limited
- Data format translation
- Error handling
- Github-specific business logic

The Client has to just connect to the server and say same language

Benefits

- N clients and M servers = M+N integrations
- No maintenance overhead
- reduce cost and time
- Better Security

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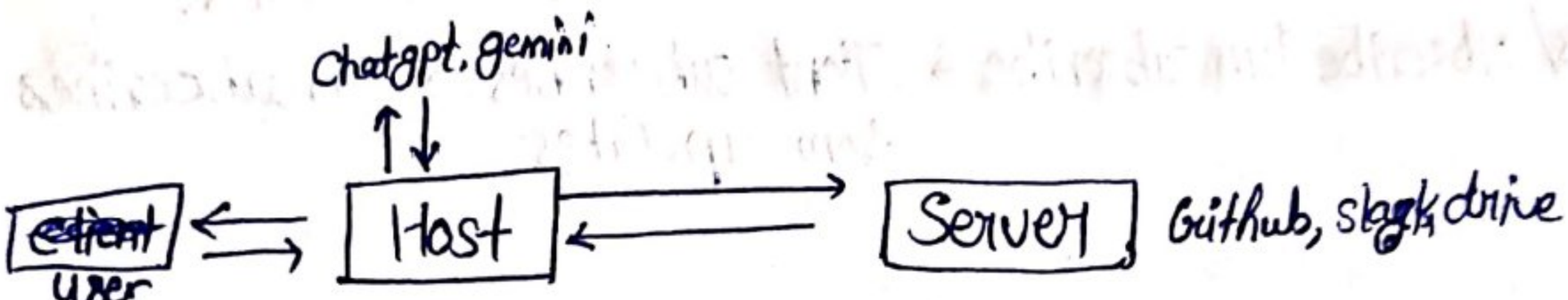
MCP Ecosystem

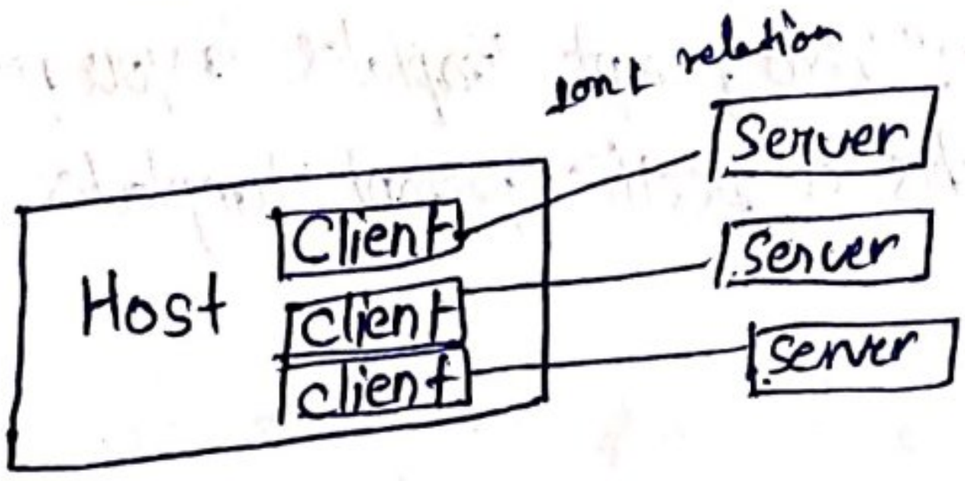
- more AI chatbots supporting MCP → more valuables for services to build MCP servers.
- More MCP server available → More valuables for AI tools to support MCP
- More adoption → more standardization → more ecosystem value
- Not supporting MCP meant being cut off for rapidly growing ecosystem.

MCP Architecture

Simple version



adding client



Benefits

Decoupling
- Safety
- Parallelism

Scalability

Primitives

- Things the server can offer to host
  - tools
  - Resources
  - Prompts.

MCP Primitives

- tools —: Actions the AI ask the server to perform.
- Resources —: Structured data sources that the AI can read.
- Prompts —: Predefined prompt templates or instructions that the server offers to help shape the AI's behavior:

Title: Login bug
Body: The login button does not work. ← normal prompts.

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Primitives - Standard operations

- Tools
  - tools/list → Client asking the server "what tools do you provide?"
  - tools/call → Client tells the server" Plz run this tool with these arguments."

- Resources
  - resources/list → Client asks" what resources are available?,"
  - resources/read → Client says" Give the content of this resources."
  - resources/subscribe/unsubscribe → Client subscribes or unsubscribes from updates

- Prompts—:
  - prompts/list → Client asks:" what prompt templates do you provide?"
  - prompts/get- Client fetches a specific prompt template.

MCP Data Layer

- Data layer- The data layer is the language and grammer of the MCP ecosystem that everyone agrees upon to communicate
- In MCP, JSON RPC 2.0 serves as the foundation of the data layer.

JSON RPC 2.0

JSON RPC stands for Javascript Object Notation - Remote Procedure Cell

- A RPC allows a program to execute a function on another computer as if it were local, hiding the details of network communication and data transfer. This abstraction masks it easier to build distributed application
- Instead of writting add(2,3) locally, you send a request to the server saying 'please run add with parameters 2 and 3!

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Request Structure

```
{
 'jsonrpc' : '2.0',
 'method' : 'add',
 'params' : [2,3],
 'id' : 1
}
```

Response Structure

```
{
 'jsonrpc' : '2.0'
 'result' : 5,
 'id' : 1
}
```

Why JSON RPC for Data layer

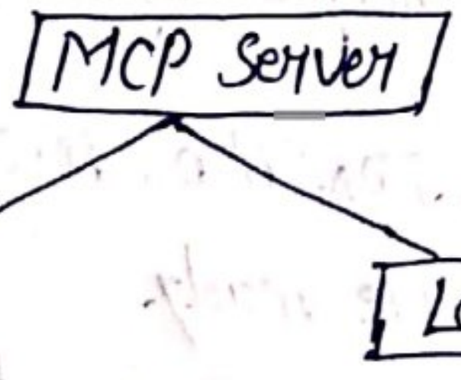
- It's light weight.
- Support by directional communication.
- It is transport-agnostic
- Supports batching.
- Supports notification.

MCP Transport Layer

The Transport Layer is the mechanism that moves JSON-RPC messages between the client and server.

- The choise of transport depends on the type of server.

MCP - Types of server



- a remote server is a prom running on another computer. that you connect to over a network.
- mode of transfer - HTTP/SSE

- a local server is a program running on your own computer.
- mode of transfer — STDIO

Local Server - STDIO

- STDIO refers to the build-in streams every program has.
  stdin (input the program read)
  stdout (output the program write)

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How does STDIO works

The host launches the server as a subprocess on the same machine.

- The host (client) writes JSON-RPC messages into the server's STDIN.
- The server reads thoes messages, processes them, and writes back responses to it's STDOUT.

Benefits of the STDIO transport

- Fast - data is passed directly b/w processes.
- Secure - no open network port that could be attacked. Communication is only local.
- simple - every language / runtime supports reading/writting from stdin/stdout. No extra libraries required.

Remote Servers — HTTP + SSE

- Using HTTP allows the host to reach Server running anywhere.
- Host sends JSON RPC requests using POST requests with a JSON payload.
- The transport supports standard HTTP authentication methords (API).
- SSE stands for Server Sent Events and its an extension of HTTP.
- Using SSE the server sends multiple msgs to client over a single open connection.
- Instead of sending one large JSON blob. the server can stream chunks of data as they are ready.
- Ideal for long running tasks or incremental updates.

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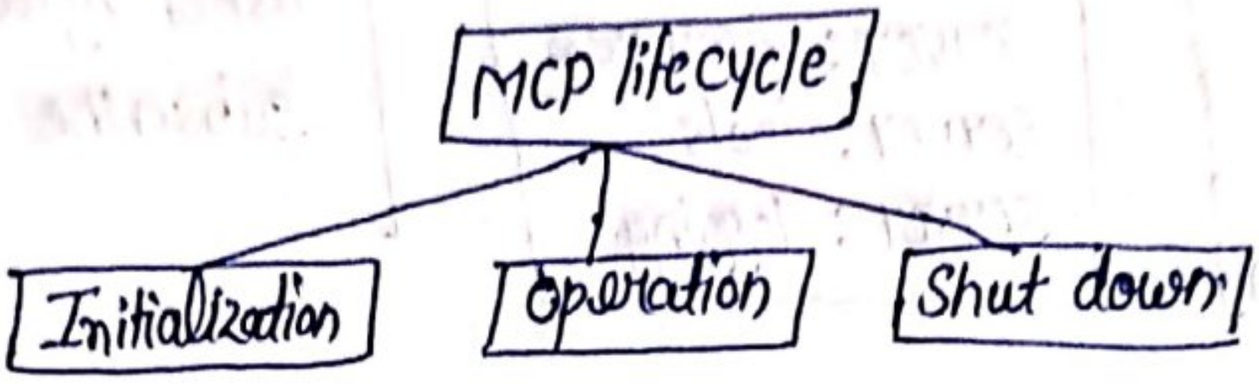
MCP life cycle.

- The MCP Life cycle describes the complete sequence of steps that govern how a host (client) and a server establish, use and end a connection during a session.



Initialization phase - The initialization phase must be the first interaction b/w client and server.

- Establish protocal version compatibility
- Exchange and negotiate capability.

Step 1: The client send a initialize request.
Step 2: The server sends its own capabilities and info.
Step 3: After successful initialization, the client MUST send an initalized notification to indicate it is ready to begin normal operations.

- Now the client and server are connected.

important Rules

- The client should not send requests other than pings before the server has responded to the intialize request.
- The server should not send requests other than pings and logging before receiving the initialized notification.

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Capability Negotiation

- Client and server capabilities establish which protocol features will be available during the session.

Client — roots Client — sampling Client — elicitation	Server : prompts Server : resources server : tools server : logging	Sub-capabilities listChanged subscribe

Operation Phase

- During the operation phase, the client and server exchange msgs according to the negotiated capabilites.
- Respect the negotiated protocol version.
- Only use capabilites that were successfully negotiated.

Shutdown Phase

- One side (typically the client) initiates shutdown.
- No special JSON RPC shutdown msg is defined.
- Transport layer is responsible for signaling termination.

Shutdown in STDIO

- Client-initiated shutdown (should):
  - Close input stream to the child process (server)
  - Wait for server to exit
  - Send SIGTERM if server does not exit in time.
  - Send SIGKILL if still ~~unle~~ unresponsive.
- Server initiated shutdown
  - Close output stream to the client
  - Exit process

shutdown in HTTP

- Client initiated shutdown (Common case):
  The client (host) closed the HTTP connection (s) it opened to the server..
- Server-initiated shutdown (possible)
  - The server may close the connection from its side.
  - The client must be prepared to detect a dropped connection and handle it (e.g. reconnect if appropriate)

Special Cases

Pings

Pings is a lightweight request/response method defined in MCP.

Purpose-: to check whether the other side (Host or Server) is still alive and connection is responsive.

When is Ping used

- Useful for checking if the otherside is up before full initialized.
- If there's no activity for a while, a client may send periodic pings. Prevents the connection from being dropped silently by the OS, proxies or fire walls.

Error Handling

- Error handling in MCP is how the Host (client) and server signal that something went wrong with a request.
- MCP inherits JSON RPC's standard error object format.

Cause of error

- Unsupported or mismatched protocol version error.
- Calling a method for a capability that wasnot negotiated
- Invalid arguments to a tool
- Internal server failure while processing a request
- Timeout exceeded → client cancels request
- Malformed JSON-RPC msgs.

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Error object structure

code - integer code that categorizes the error.
message - short human-readable explanation.
data - (optional) extra structured data for debugging or context.

Common Error Codes

Error Code	Name
-32601	Method not found
-32602	Invalid parameters
-32600	Invalid request
-32700	Parse error
-32000 or above	Server defined error

Time Out

Timeout is about ensuring request don't hang forever.

Purpose -
- Protects against unresponsive or overload servers.
- Ensures resources (memory, CPU) aren't held indefinitely
- Gives the user feedback instead of waiting forever.

How timeout works
- SDKs let client sets a per-request timeout (e.g. 30s)
- If the deadline passes with no result → client triggers a timeout.
- Client then sends a cancellation notification to tell the server to stop.
- The server must stop processing that request and not return a result.

Progress Notification
- Purpose: Let the client know a long-running request is still making progress.
- client includes a progress token in the request's _meta.
- server can then send notification / progress update while working.

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Connectors

- A connector is a built in feature that links claude to MCP server automatically, without the need for manual setup or Configuration
- Most claude Desktop users are non technical end-users who just want Claude to talk to their apps (Notion, Google drive, Github)
- They do not want to run servers, edit JSON or worry about transports.
- The connector system wraps an MCP server behind the scenes and handles authentication via OAuth (Sign in with google, GitHub)
- This keeps things easy, safe and consistent.
- Think of connectors as the 'Appstore' for MCP server - user-friendly click-based, pre curated.

Why not use Connectors always?

- Connectors Are curated & manged.
  - Connector are officially built, hosted and maintained by Anthropic
  - They come with oauth login flows, managed security, rate limits and guaranteed stability.
- MCP is an open standard
  - MCP is designed so anyone can write a server.
  - Forcing everthing through Connectors would close the ecosystem and make you dependent on Anthropic to approve or publish server.

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Fast MCP or MCP SDK

1. Birth of MCP (2023-2024)
   - Antropic introduces Model Context Protocol (MCP)
     Structured communication b/w LLM & external tools / data.
   - Early needs:
     - build MCP server (tools / resources / prompts)
     - build MCP clients (apps consuming those servers)
   - Official python SDK (mcp) released
     - mcp.server → build servers
     - mcp.client → build clients
     - mcp.cli → CLI for debugging / testing
   - This becomes the reference implementation

2. Develop friction & FastMCP 1.x (2024)
   - Problem: Raw SDK = verbose & boilerplate heavy
     - manual List_tools, call_tool, list_prompts, etc.
     - Explict schema definitions required
   - Solution: FastMCP helper inside SDK (mcp.server.fastMCP)
   - Decorator-driven, type hint aware.

```
from mcp.server.fastmcp import FastMCP
mcp = FastMCP('Demo')
@mcp.tool()
def add(a:int, b:int) → int:
    return a+b
```

3. ~~MCP~~ Fast MCP Break Out (2025)
   - FastMCP adoption grow rapidly
   - SDK maintain's goles:
     - Keep SDK minimal & spec foucsed
     - Let FastMCP evolve faster (CQol APIs, transports, integrations)

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- FastMCP 2.x released as standalone packages.

4. Where Things Stand (Late 2025)

- MCP SDK
  - official. spec-compliant. stable
  - install: pip install mcp[cli]

- MCP CLI (mcp--)
  - ships with SDK
  - Inspect debug. test servers
- FastMCP 2.x (fast. mcp)
  - standalone, community-driven. fast-evolving
  - Ideal for students educators. rapid prototypes
  - Not offical reference. but more ergonomic

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